

Supplementary Information

Liquid metal dispersed single-atom catalyst with high-temperature stability

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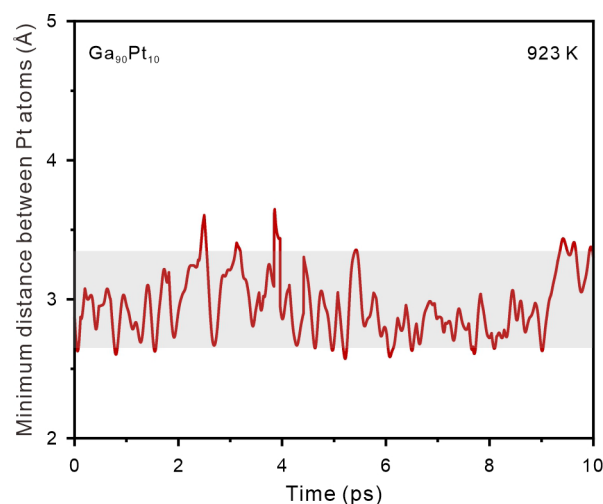
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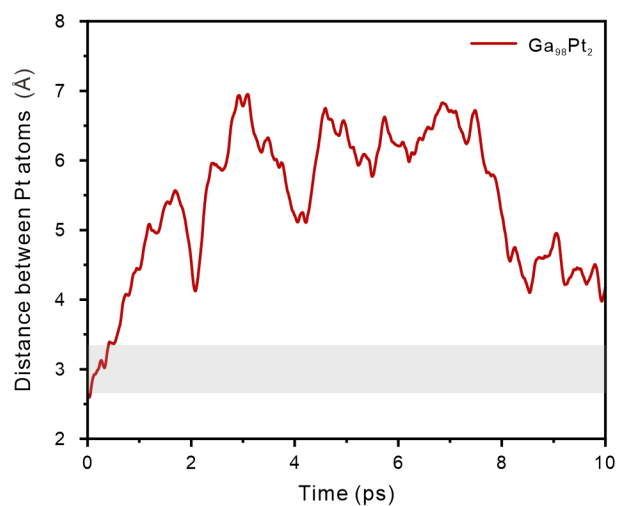
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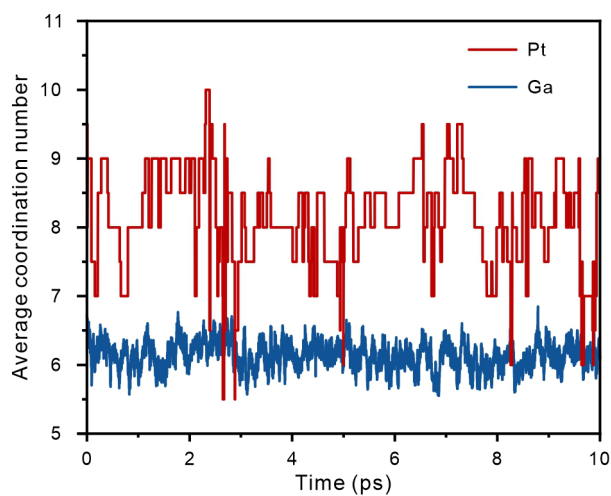
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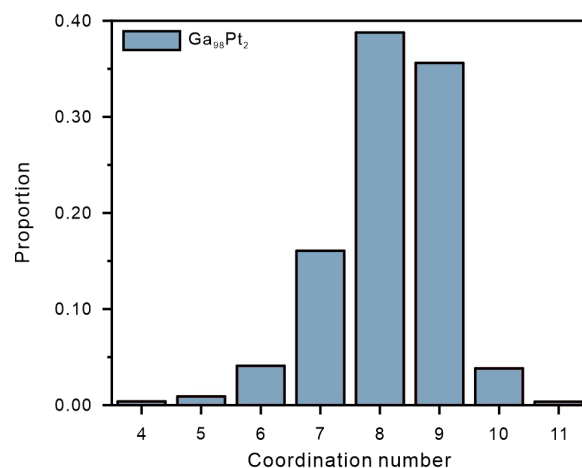
Supplementary Fig. 1 | Time evolution of the minimum Pt–Pt distance in the Ga₉₀Pt₁₀ system at 923 K. The gray area marked is 2.65 ~ 3.35 Å, which is 1.2 times the length of the Pt–Pt bond (2.65 ~ 2.79 Å). In the Ga₉₀Pt₁₀ system, the shortest Pt–Pt distance fluctuates between 2.51 Å and 3.65 Å, which are partly in the range of 1.2 times the length of the Pt–Pt bond (2.65 ~ 2.79 Å)¹. Generally, when the distance between the two atoms is less than 1.2 times (2.79 Å × 1.2 = 3.35 Å) of the sum of their Cambridge Structural Database (CSD) covalent radii, the chemical bond between the two atoms can be considered not to be formed^{2,3} and they are isolated from each other. The relatively high Pt concentration in the Ga₉₀Pt₁₀ system leads to intermittent formation of the Pt₂ dimer.



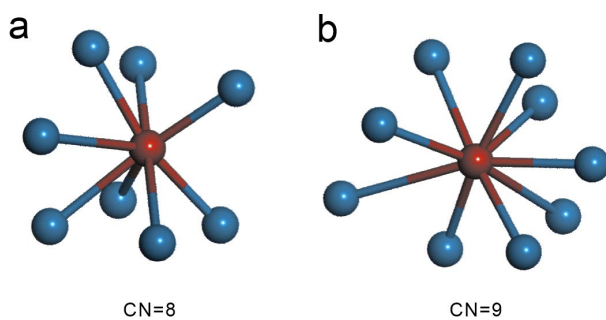
Supplementary Fig. 2 | Time evolution of Pt–Pt interatomic distance in the $\text{Ga}_{98}\text{Pt}_2$ system under NPT ensemble simulation. The final structure from the simulation was used as the initial structure for the following NVT ensemble simulation.



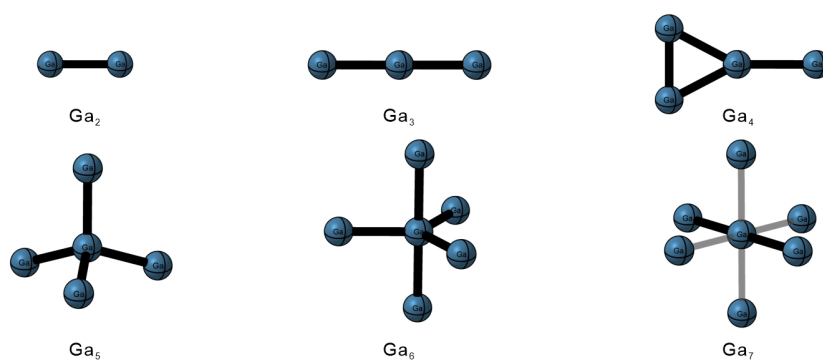
Supplementary Fig. 3 | Variation of the average coordination numbers of Pt and Ga per frame with simulation time in the Ga₉₈Pt₂ system at 923 K. The simulation results indicate that at 923 K, the coordination number (CN) of Ga is predominantly 6, while that of Pt mainly falls within the range of 7 ~ 9.



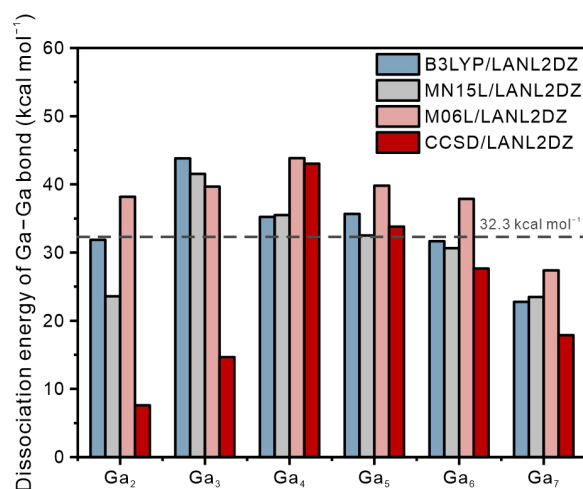
Supplementary Fig. 4 | Statistical distribution of Pt coordination numbers in Ga₉₈Pt₂ from AIMD simulations. The coordination environment of Pt atoms in the Ga₉₈Pt₂ system shows that the dominant coordination numbers for individual Pt atoms are 8 or 9, implying that mono-dispersed Pt atoms migrate within liquid Ga in the form of clusters such as Ga₈Pt or Ga₉Pt.



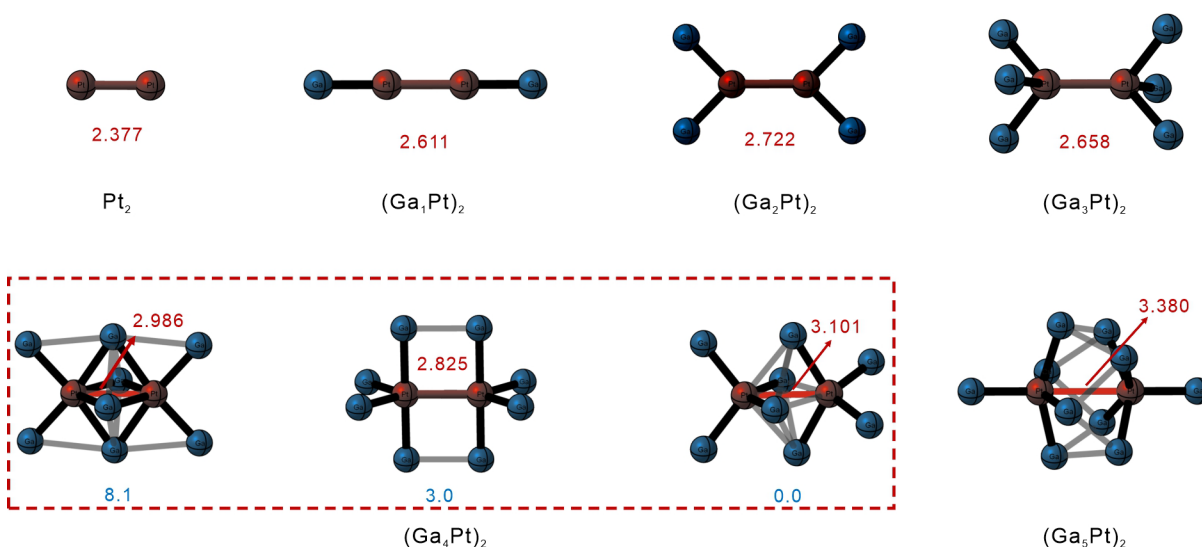
Supplementary Fig. 5 | Structures of the Ga₈Pt (a) and Ga₉Pt (b) clusters. The spontaneous dispersion of Pt into Ga₈Pt and Ga₉Pt clusters occurs in the presence of an abundance of Ga atoms.



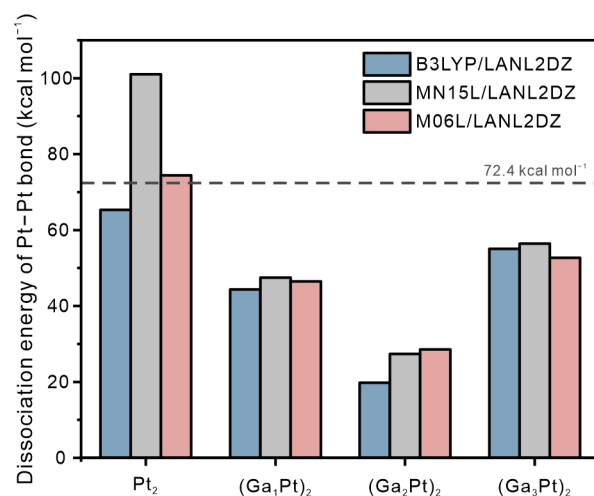
Supplementary Fig. 6 | The optimized structures of the Ga_2 , Ga_3 , Ga_4 , Ga_5 , Ga_6 , and Ga_7Pt clusters.



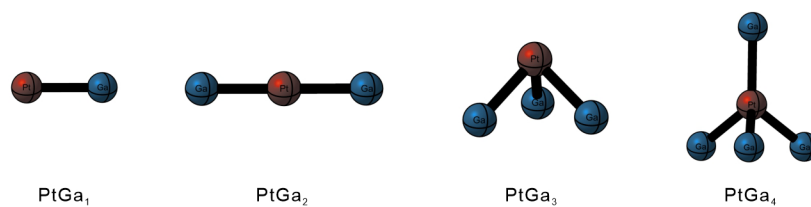
Supplementary Fig. 7 | The calculated dissociation energies of Ga–Ga bond in Ga₂, Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇ clusters obtained using the B3LYP, MN15L, M06L, and CCSD methods. The Ga–Ga bond energies in the Ga₂ molecule calculated using the B3LYP, MN15L, M06L, and CCSD methods are 31.9, 23.6, 38.2, and 7.6 kcal mol⁻¹, respectively. Among these, the B3LYP result (31.9 kcal mol⁻¹) is closest to the experimental value⁴ of 32.3 kcal mol⁻¹, while the CCSD result deviates significantly from the experimental value, clearly indicating that the CCSD method is not suitable for simulating Ga–Pt systems. Furthermore, the average Ga–Ga bond energies in Ga₂, Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇ clusters calculated using B3LYP, MN15L, and M06L methods are 33.5, 31.2, and 37.8 kcal mol⁻¹, respectively, demonstrating consistency among these three methods.



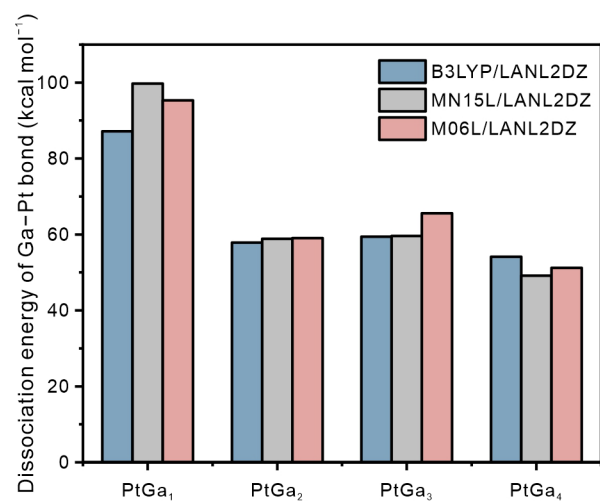
Supplementary Fig. 8 | The optimized structures of the molecular systems Pt_2 , $(\text{Ga}_1\text{Pt})_2$, $(\text{Ga}_2\text{Pt})_2$, $(\text{Ga}_3\text{Pt})_2$, $(\text{Ga}_4\text{Pt})_2$, and $(\text{Ga}_5\text{Pt})_2$. The red numbers denote the Pt-Pt bond lengths in angstroms (Å), and the blue numbers denote the relative energies (in kcal mol^{-1}) of the three isomers of $(\text{Ga}_4\text{Pt})_2$. Based on the optimized geometries, the Pt-Pt bond lengths in the stable structures of $(\text{Ga}_4\text{Pt})_2$ and $(\text{Ga}_5\text{Pt})_2$ are 3.101 Å and 3.380 Å, respectively. These values significantly exceed 1.2 times the Pt-Pt bond length of 2.377 Å in Pt_2 , indicating that the Pt-Pt bonds in the stable configurations of $(\text{Ga}_4\text{Pt})_2$ and $(\text{Ga}_5\text{Pt})_2$ can be considered effectively broken. Therefore, we calculated the bond energy difference ΔE_b only for the Pt_2 , $(\text{Ga}_1\text{Pt})_2$, $(\text{Ga}_2\text{Pt})_2$, and $(\text{Ga}_3\text{Pt})_2$ molecular clusters.



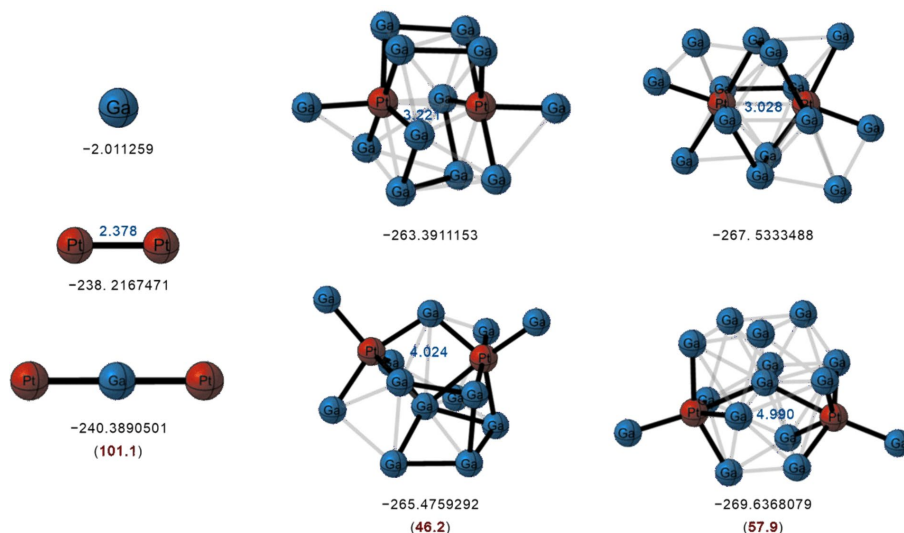
Supplementary Fig. 9 | The calculated dissociation energies of Pt–Pt bond in Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ clusters obtained using the B3LYP, MN15L, M06L methods. the Pt–Pt bond energies in (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ clusters calculated using B3LYP, MN15L, and M06L methods are similar, while the Pt–Pt bond energies in the Pt₂ molecule calculated with these methods are 65.3, 101.0, and 74.4 kcal mol⁻¹, respectively. Among these, the B3LYP and M06L results are close to the experimental value of 72.4 kcal mol⁻¹[5].



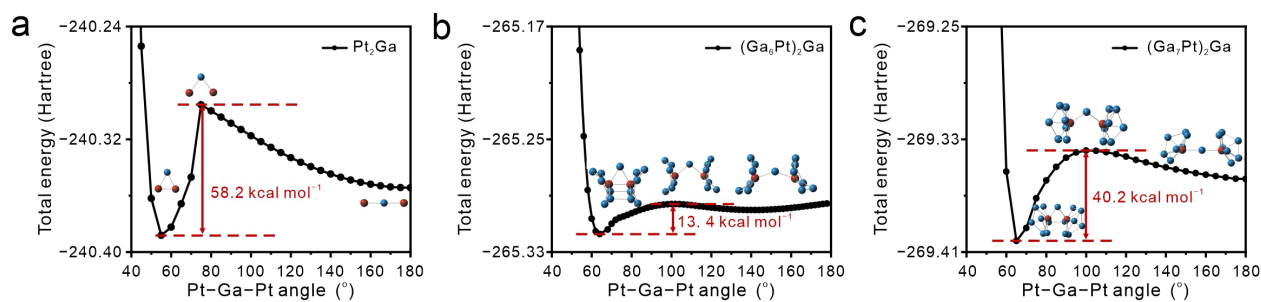
Supplementary Fig. 10 | The optimized structures of the molecular systems PtGa₁, PtGa₂, PtGa₃, and PtGa₄.



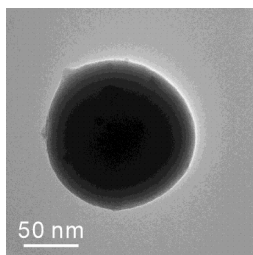
Supplementary Fig. 11 | The calculated dissociation energies of Ga–Pt bond in PtGa₁, PtGa₂, PtGa₃, and PtGa₄ clusters obtained using the B3LYP, MN15L, M06L methods.



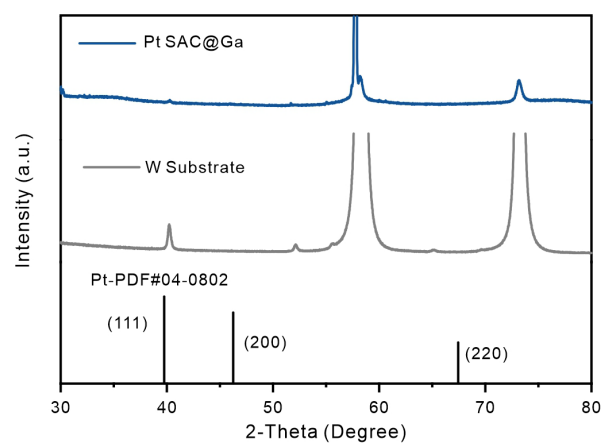
Supplementary Fig. 12 | The formation energies of Pt₂Ga, (Ga₆Pt)₂Ga, and (Ga₇Pt)₂Ga molecules. The figure presents the optimized structures of the molecular systems Pt₂, (Ga₆Pt)₂, (Ga₇Pt)₂, Pt₂Ga, (Ga₆Pt)₂Ga, and (Ga₇Pt)₂Ga. The numbers labeled in black represent the energies of the systems, with the unit in Hartree; the numbers labeled in blue denote bond lengths, with the unit in Å. The numbers labeled in red indicate the calculated formation energies ($\Delta E = E_{\text{Ga}} + E_{\text{Pt-Ga molecule}} - E_{\text{complex}}$), with the unit in kcal mol⁻¹. The formation energies of Pt₂Ga, (Ga₆Pt)₂Ga, and (Ga₇Pt)₂Ga molecules were determined to be 101.1, 46.2, and 57.9 kcal mol⁻¹, respectively. These results thermodynamically demonstrate that Ga can readily disrupt the Pt–Pt bonds, leading to the formation of single-atom Pt in liquid Ga.



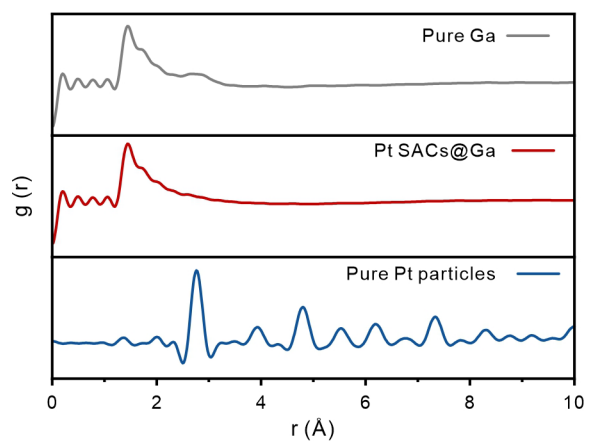
Supplementary Fig. 13 | The energy barriers of Ga atom insertion into the Pt–Pt bond by performing rigid scans of the Pt–Ga–Pt angle. The energy barriers of Ga atom insertion into the Pt–Pt bond in Pt_2 (a), $(\text{Ga}_6\text{Pt})_2$ (b), and $(\text{Ga}_7\text{Pt})_2$ (c) were calculated to be 58.2, 13.4, and 40.2 kcal mol^{−1}, respectively. The energy barrier for Ga inserting into the Pt–Pt bond in the $(\text{Ga}_7\text{Pt})_2\text{Ga}$ system, though higher than that in $(\text{Ga}_6\text{Pt})_2\text{Ga}$, is still significantly lower than the typical ~67 kcal mol^{−1} barrier that chemical reactions can generally overcome at 923 K. This indicates that the $(\text{Ga}_7\text{Pt})_2\text{Ga}$ system is kinetically feasible, thereby further facilitating the formation of Ga_8Pt clusters.



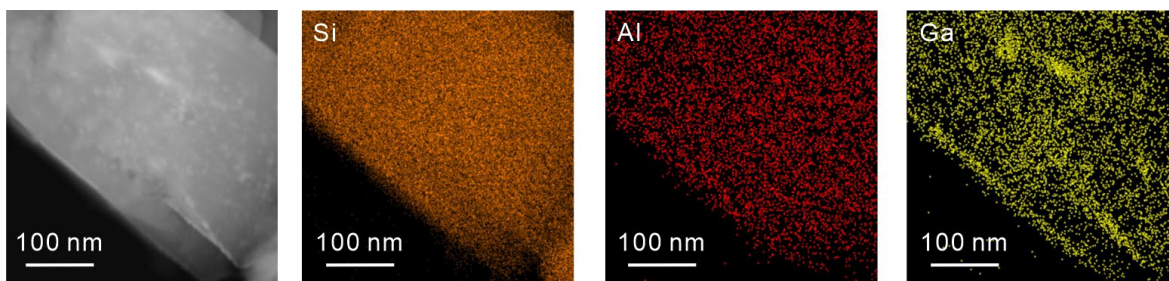
Supplementary Fig. 21 | Scanning transmission electron microscopy (STEM) image of Pt SACs@Ga.



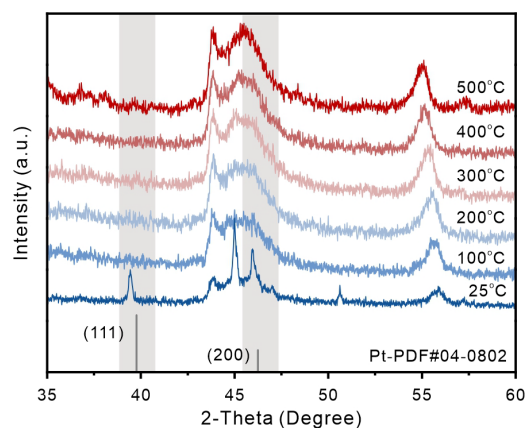
Supplementary Fig. 22 | X-ray diffraction (XRD) patterns of Pt in Ga. Three sets of peaks appeared at 39.8°, 46.2°, and 67.5°, which are ascribed to the (111), (200), and (220) reflections of Pt (PDF # 04–0802). The peaks of the Pt metal were absent in the liquid metal, indicating there was no obvious agglomeration of Pt.



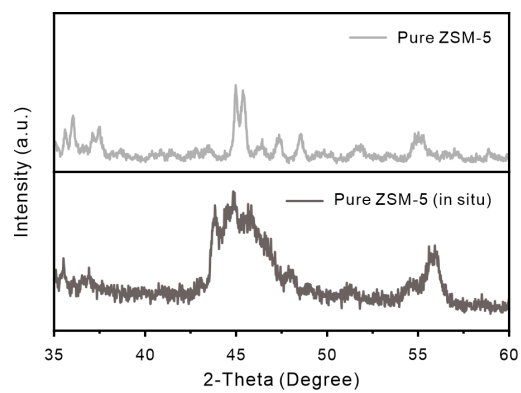
Supplementary Fig. 23 | Pair distribution function (PDF) for Pt SACs@Ga. The PDF spectra for pure Pt particles and pure Ga were included for comparison. The PDF clearly demonstrates that the Pt species are in isolated environments without Pt–Pt bond.



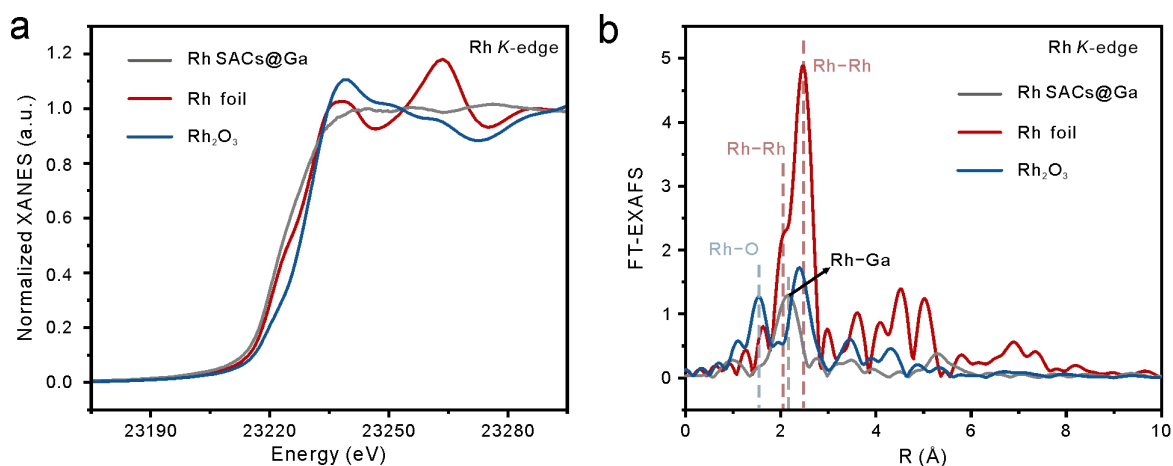
Supplementary Fig. 24 | The STEM image and EDS mappings of Ga loaded on ZSM-5 zeolites. Ga was loaded on the ZSM-5 zeolites to increase the mass transfer of gas in the catalyst.



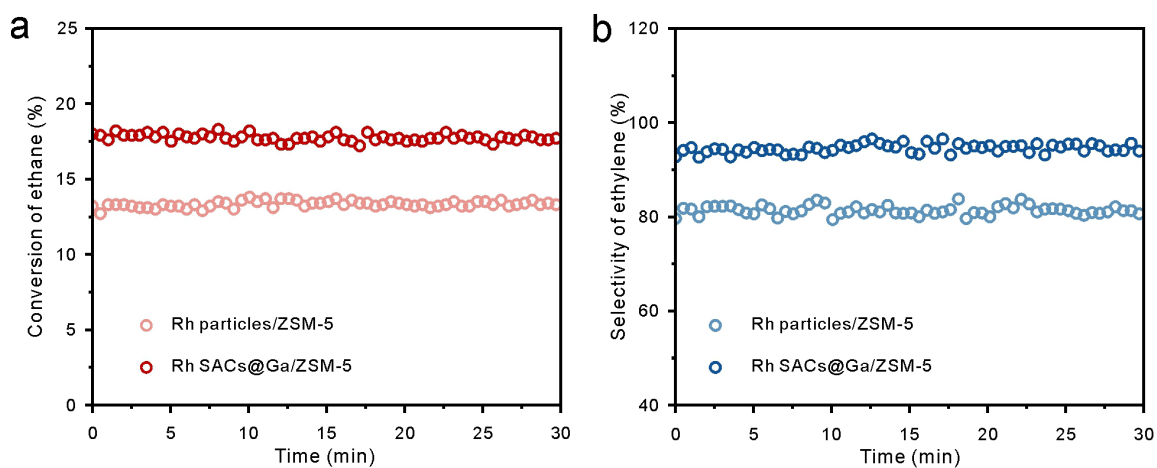
Supplementary Fig. 25 | In situ XRD patterns of Pt SACs@Ga/ZSM-5 in the process of heating. Initially, the characteristic peaks of Pt metals can be apparently observed, which are located at 39.8° and 46.3° . With the temperature increasing, the characteristic peaks of Pt metals were gradually weakened and subsequently disappeared.



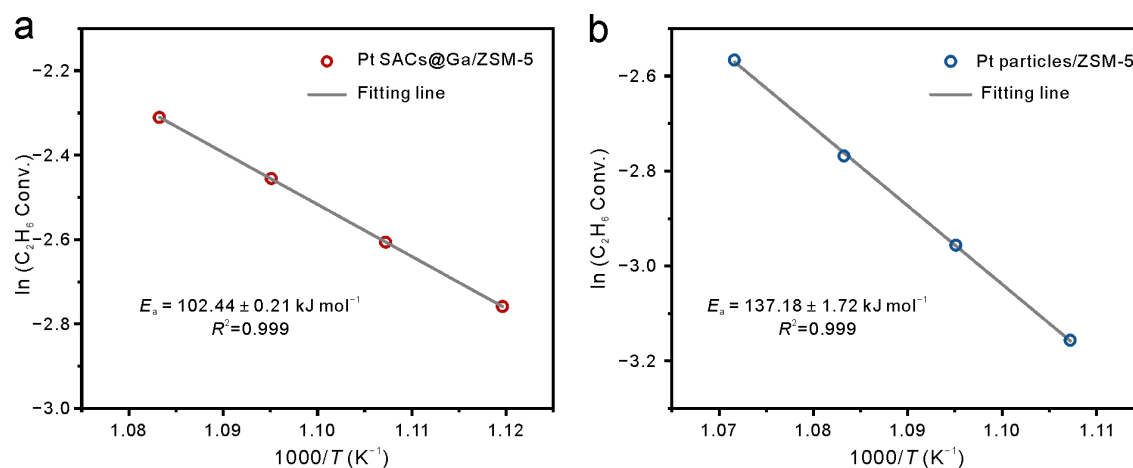
Supplementary Fig. 26 | XRD patterns of pure ZSM-5 and ZSM-5 in in situ equipment. The wide diffraction peak observed at $43^{\circ} \sim 47^{\circ}$ originates from the signal of the carbon within the in situ reactor.



Supplementary Fig. 27 | Characterization of liquid metal dispersed Rh SACs. (a) Normalized XANES spectra at the Rh K-edge. (b) FT Rh K-edge EXAFS spectra in R space of liquid catalyst with phase correction. When Rh particles were dispersed in the liquid metal in the same approach, the formation of Rh–Rh bonds at 2.45 Å and 2.05 Å was not observed, nor was the formation of Rh–O bonds at 1.51 Å detected. Instead, a unique new bond was observed at the 2.15 Å position, which may be attributed to the formation of Rh–Ga bond. Synchrotron radiation data clearly demonstrates the formation of Rh single atoms in liquid metal.



Supplementary Fig. 28 | Ethane dehydrogenation performance of liquid metal dispersed Rh SACs. (a) Conversion of ethane. (b) Selectivity of ethylene.



Supplementary Fig. 29 | Arrhenius plots for EDH reaction. (a) Pt SACs@Ga/ZSM-5. (b) Pt particles/ZSM-5. To investigate the kinetic behavior of the catalyst for ethane dehydrogenation, we measured reaction rates under conditions designed to minimize transport limitations. Specifically, we limited the conversion to 5 ~ 15% while ensuring ethylene selectivity remained above 90% by systematically varying the catalyst amount. However, due to the poor selectivity of Pt particles/ZSM-5, side reactions will occur inevitably. Therefore, we only controlled its conversion rate to be within the range of 5 ~ 15%.

Supplementary Tab. 1 | Comparison of thermal stability in this work and other SACs among reported works.

Catalyst	Reaction types	Temperature (°C)	Time (h)	Reference
Pt SACs@Ga/ZSM-5	Ethane dehydrogenation	650	100	This work
Ce-MoTiO _x	Reduction of NO _x with NH ₃	300	240	7
Rh ₁ /Fe-doped CeO ₂	Reduction of NO with CO	400	100	8
Co@MFI	Non-oxidative dehydrogenation of ethane	650	40	9
In/Rh@S-1	Propane dehydrogenation	600	5500	10
Fe(II)-S-1	Ethane dehydrogenation	550	6	11
Co/S-1	Ethane dehydrogenation	590	102	12
Ni/A-TiO ₂	Propane dehydrogenation	580	5	13
GaPt/ZSM-5	Oxidative ethane dehydrogenation	650	3	14
Ni/ α -MoC	Methanol dehydrogenation	240	100	15
Pt _x /Ga	Hydrogenation of acid to alkanes	255	6	16
Ni ₁ -Mg SAA	Mg hydrogen storage reaction	300	30 cycles	17
Ni SAs/N-Al ₂ O ₃	Propane dehydrogenation	580	140	18
Cu-Fe ₂ O ₃	NH ₃ oxidation	300	100	19
MOF-808-NaCu	CO ₂ hydrogenation to methanol	275	40	20
Co-Ba/Y ₂ O ₃	Ammonia decomposition	500 ~ 600	110 h at 550 °C and 240 h at 500 °C	21
Pt/CeO _x /SiO ₂	H ₂ reduction	500	12	22
Cu ₁ / γ -Al ₂ O ₃	Methane to methanol	200	12	23
Ru-(H)L(i-C ₃) ₂ /AC	Acetylene hydrochlorination	170	500	24
RuNi SAA	Hydrogenation of nitroarenes	60	3	25
Pd ₁ Ni/SiO ₂	Hydrogenation of acetylene	80	12	26
Ru ₁ CoNP	Synthesis of piperidines and pyridine from furfural	220	4	27
RuNi SAA	CO ₂ hydrogenation	400	140	28
Ni ₁ /CoCeO ₂	Dry reforming of methane	800	150	29
Co ₁ /V ₂ O ₅	Acceptorless dehydrogenative coupling of benzyl alcohol	359	0.5	30
Pt-NiCe@SiO ₂	Dry reforming of methane	550	120	31
Mn/V/Al ₂ O ₃ and Mn/Ni/ZrO ₂	Oxidative dehydrogenation of propane	300 ~ 500	25	32
Fe-ZIF-8-800	Hydrogenation of furfural	60 ~ 120	6	33
Rh ₁ /SnO ₂	Ethylene hydroformylation	150	85	34
Mo/NiFe ₇₁ -V	Hydrodeoxygenation of lignin to naphthene	240	3	35
Ni-SAs/N-CNTs	Reverse water-gas shift reaction	500	100	36
Co _{0.11} %@0.08Mo-S-1	Propane dehydrogenation	600	72	37
PtSn/nano-Al ₂ O ₃	Propane dehydrogenation	600	40	38
B-Ni in MnO ₂	Methane catalytic combustion	500	32	39

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